



DP-601 : Implement a Lakehouse with Microsoft Fabric

Exercise 2 : Analyze data with Apache Spark in Fabric

- Create a workspace
- Create a lakehouse and upload files
- Create a notebook
- Create a DataFrame
- Explore data in a DataFrame
- Use Spark to transform data files
- Work with tables and SQL
- Visualize data with Spark



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Create a workspace

Create a workspace

×

Name *

Apache_spark

✓ This name is available

Description

Describe this workspace

Domain ⓘ

Assign to a domain (optional) ▾

Learn more about workspace settings

[↗](#)

Workspace image



↑ Upload

↶ Reset

Advanced ^

Contact list * ⓘ

 dagstudentsgreatlearning (Owner) ×

Workspace type ⓘ

What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#) [↗](#)

— Power BI only workspace types

Power BI workspaces only support the use of Power BI capabilities.

Apply

Cancel



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➤ Create a lakehouse and upload files

New Lakehouse

Name *

Location *

 MS Fabric Practice ▼ ↗

☒ Lakehouse schemas ⓘ

CreateCancel

Explorer

Add lakehouses

Spark_LH

Tables

dbo

Files

Files

MS Fabric Practice > Spark_LH > Files

Showing 3 items

Search files

Name	Date modified	Type	Size
 2019.csv	6/12/2026, 4:09:01 ...	csv	120 KB
 2020.csv	6/12/2026, 4:09:02 ...	csv	279 KB
 2021.csv	6/12/2026, 4:09:28 ...	csv	2 MB



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Create a notebook



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Create a DataFrame

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PySpark (Python)

Environment

Workspace default

Data Wrangler

Copilot

Explorer

Data items

Resources

Connections

OneLake

Spark_LH

Tables

Files

Files

2019.csv

2020.csv

2021.csv

Sales order data exploration

Use this notebook to explore sales order data

1 df = spark.read.format("csv").option("header","false").load("Files/2019.csv")

2 # df now is a Spark DataFrame containing CSV data from "Files/2019.csv".

3 display(df)

2 sec - Command executed in 2 sec 356 ms by DAG on 6/15/2026, 12:41:05 AM

Spark jobs (2 of 2 succeeded)

Resources

Log

Table

New chart

Data Wrangler

Table view

	ABC_c0	ABC_c1	ABC_c2	ABC_c3	ABC_c4	ABC_c5	ABC_c6	ABC_c7	ABC_c8
1	SO43701	1	2019-07-01	Christy Zhu	christy12@...	Mountain-...	1	3399.99	271.9992
2	SO43704	1	2019-07-01	Julio Ruiz	julio1@adv...	Mountain-...	1	3374.99	269.9992
3	SO43705	1	2019-07-01	Curtis Lu	curtis9@ad...	Mountain-...	1	3399.99	271.9992
4	SO43700	1	2019-07-01	Ruben Pras...	ruben10@...	Road-650 ...	1	699.0982	55.9279
5	SO43703	1	2019-07-01	Albert Alva...	albert7@a...	Road-150 ...	1	3578.27	286.2616
6	SO43697	1	2019-07-01	Cole Watson	cole1@adv...	Road-150 ...	1	3578.27	286.2616
7	SO43699	1	2019-07-01	Sydney Wri...	sydney61...	Mountain-...	1	3399.99	271.9992
8	SO43702	1	2019-07-01	Colin Anand	colin45@a...	Road-150 ...	1	3578.27	286.2616
9	SO43698	1	2019-07-01	Rachael M...	rachael16...	Mountain-...	1	3399.99	271.9992
10	SO43707	1	2019-07-02	Emma Brown	emma3@a...	Road-150 ...	1	3578.27	286.2616
11	SO43711	1	2019-07-02	Courtney E...	courtney1...	Road-150 ...	1	3578.27	286.2616
12	SO43706	1	2019-07-02	Edward Bro...	edward26...	Road-150 ...	1	3578.27	286.2616
13	SO43708	1	2019-07-02	Brad Dang...	brad23@ad...	Road-650 ...	1	699.0982	55.9279



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Create a DataFrame

HomeEditAI toolsRunView

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Run allStandard sessionPySpark (Python)EnvironmentWorkspace defaultData WranglerCopilot

Explorer

```
1 from pyspark.sql.types import *
2
3 orderSchema = StructType([
4     StructField("SalesOrderNumber", StringType()),
5     StructField("SalesOrderLineNumber", IntegerType()),
6     StructField("OrderDate", DateType()),
7     StructField("CustomerName", StringType()),
8     StructField("Email", StringType()),
9     StructField("Item", StringType()),
10    StructField("Quantity", IntegerType()),
11    StructField("UnitPrice", FloatType()),
12    StructField("Tax", FloatType())
13 ])
14
15 df = spark.read.format("csv").schema(orderSchema).load("Files/2019.csv")
16
17 display(df)
```

[5] ✓ 4 sec - Command executed in 4 sec 663 ms by DAG on 6/15/2026, 12:50:59 AM

Spark jobs (1 of 1 succeeded)ResourcesLog

TableNew chart

Data Wrangler9 columns, 1000 rows

DownloadFilter by keyword

	ABC SalesOrderNumber	123 SalesOrderLineNumber	OrderDate	ABC CustomerName	ABC Email	ABC Item	123 Quantity	12F UnitPrice	12F Tax
1	SO43701	1	2019-07-01	Christy Zhu	christy12@...	Mountain-...	1	3399.99	271.9992
2	SO43704	1	2019-07-01	Julio Ruiz	julio1@adv...	Mountain-...	1	3374.99	269.9992
3	SO43705	1	2019-07-01	Curtis Lu	curtis9@ad...	Mountain-...	1	3399.99	271.9992
4	SO43700	1	2019-07-01	Ruben Prasad	ruben10@...	Road-650 ...	1	699.0982	55.9279
5	SO43703	1	2019-07-01	Albert Alvarez	albert7@a...	Road-150 ...	1	3578.27	286.2616
6	SO43697	1	2019-07-01	Cole Watson	cole1@adv...	Road-150 ...	1	3578.27	286.2616

Inspect

Session readyAutoSave: OnSelected Cell 3 of 3



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➤ Explore data in a DataFrame

Copilot

Create

Browse

Apps

Scorecards

OneLake catalog

Learn

Workspaces

MS Fabric Practice

Notebook 2

Power BI

»

▶ ▼

Explorer

```
1 customers= df.select('CustomerName','Email')
2
3 print(customers.count())
4 print(customers.distinct().count())
5
6 display(customers.distinct())
```

[14] ✓ 3 sec - Command executed in 3 sec 550 ms by DAG on 6/15/2026, 1:01:34 AM

> Spark jobs (7 of 7 succeeded) Resources Log

... 32718
12427

Table

+ New chart

Table view

	ABC CustomerName	ABC Email
1	Bridget Andersen	bridget15...
2	Mya Butler	mya14@ad...
3	Deanna Hernandez	deanna29...
4	Ricky Navarro	ricky10@a...
5	Omar Ye	omar9@ad...
6	Kellie Gutierrez	kellie9@ad...
7	Raymond Rana	raymond13...
8	Derrick Moreno	derrick6@a...
9	Megan Walker	megan25...
10	Edward Jackson	edward34...
11	Wayne She	wayne1@a...
12	Willie Shan	willie29@a...
13	Jake Guo	jake15@ad...

✓ Session ready

AutoSave: On



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➤ Explore data in a DataFrame

```
1 ProductSales=df.select("Item","Quantity").groupBy('Item').sum()  
2 display(ProductSales)
```

[10] ✓ 2 sec - Command executed in 2 sec 143 ms by DAG on 6/15/2026, 11:32:28 AM

> Spark jobs (2 of 2 succeeded) Resources Log

...

Table New chart

Table view

	ABC Item	12L sum(Quantity)
1	Mountain-...	388
2	Touring-10...	74
3	Touring-10...	67
4	Short-Sleev...	216
5	Women's ...	146
6	Long-Sleev...	234
7	Mountain-...	59
8	Racing Soc...	165
9	Mountain-...	359
10	Road-750 ...	170

```
1 from pyspark.sql.functions import *  
2  
3 yearlySales = df.select(year(col("OrderDate")).alias("Year")).groupBy("Year").count().orderBy("Year")  
4  
5 display(yearlySales)
```

✓ 1 sec - Command executed in 1 sec 533 ms by DAG on 6/15/2026, 11:46:21 AM

> Spark jobs (2 of 2 succeeded) Resources

Table New chart

Table view

	123 Year	12L count
1	2019	1201
2	2020	2733
3	2021	28784



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Use Spark to transform data files

1 from pyspark.sql.functions import *
2
3 # Create Year and Month columns
4 transformed_df = df.withColumn("Year", year(col("OrderDate"))).withColumn("Month", month(col("OrderDate")))
5
6 # Create the new FirstName and LastName fields
7 transformed_df = transformed_df.withColumn("FirstName", split(col("CustomerName"), " ").getItem(0)).withColumn("LastName", split(col("CustomerName"), " ").getItem(1))
8
9 # Filter and reorder columns
10 transformed_df = transformed_df[["SalesOrderNumber", "SalesOrderLineNumber", "OrderDate", "Year", "Month", "FirstName", "LastName", "Email", "Item", "Quantity", "UnitPrice"]]
11
12 # Display the first five orders
13 display(transformed_df.limit(5))

✓ 1 sec - Command executed in 1 sec 525 ms by DAG on 6/15/2026, 11:55:54 AM

PySpark (Python)

Spark jobs (1 of 1 succeeded) Resources Log

Table + New chart

12 columns, 5 rows

Table view

Download Filter by keyword

	ABC SalesOrderNumber	123 SalesOrderLineNumber	OrderDate	123 Year	123 Month	ABC FirstName	ABC LastName	ABC Email	ABC Item	123 Quantity	12F UnitPrice
1	SO49171	1	2021-01-01	2021	1	Mariah	Foster	mariah21@...	Road-250 ...	1	2181.5625
2	SO49172	1	2021-01-01	2021	1	Brian	Howard	brian23@a...	Road-250 ...	1	2443.35
3	SO49173	1	2021-01-01	2021	1	Linda	Alvarez	linda19@a...	Mountain-...	1	2071.4197
4	SO49174	1	2021-01-01	2021	1	Gina	Hernandez	gina4@adv...	Mountain-...	1	2071.4197
5	SO49178	1	2021-01-01	2021	1	Beth	Ruiz	beth4@ad...	Road-550-...	1	1000.4375

Inspect

1 from pyspark.sql.functions import *
2
3 # Create Year and Month columns
4 transformed_df = df.withColumn("Year", year(col("OrderDate"))).withColumn("Month", month(col("OrderDate")))
5
6 # Create the new FirstName and LastName fields
7 transformed_df = transformed_df.withColumn("FirstName", split(col("CustomerName"), " ").getItem(0)).withColumn("LastName", split(col("CustomerName"), " ").getItem(1))
8
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10 transformed_df = transformed_df[["SalesOrderNumber", "SalesOrderLineNumber", "OrderDate", "Year", "Month", "FirstName", "LastName", "Email", "Item", "Quantity", "UnitPrice"]]
11
12 # Display the first five orders
13 display(transformed_df.limit(5))

✓ 1 sec - Command executed in 1 sec 525 ms by DAG on 6/15/2026, 11:55:54 AM

PySpark (Python)

Spark jobs (1 of 1 succeeded) Resources Log

Table + New chart

12 columns, 5 rows

Table view

Download Filter by keyword

	ABC SalesOrderNumber	123 SalesOrderLineNumber	OrderDate	123 Year	123 Month	ABC FirstName	ABC LastName	ABC Email	ABC Item	123 Quantity	12F UnitPrice
1	SO49171	1	2021-01-01	2021	1	Mariah	Foster	mariah21@...	Road-250 ...	1	2181.5625
2	SO49172	1	2021-01-01	2021	1	Brian	Howard	brian23@a...	Road-250 ...	1	2443.35
3	SO49173	1	2021-01-01	2021	1	Linda	Alvarez	linda19@a...	Mountain-...	1	2071.4197
4	SO49174	1	2021-01-01	2021	1	Gina	Hernandez	gina4@adv...	Mountain-...	1	2071.4197

Inspect



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► Work with tables and SQL

```
1 df.write.format("delta").saveAsTable("salesorders")
2 spark.sql("DESCRIBE EXTENDED salesorders").show(truncate=False)
```

[154] ✓ 15 sec - Command executed in 15 sec 896 ms by DAG on 6/15/2026, 12:25:42 PM

> Spark jobs (6 of 6 succeeded) Resources

...

col_name	data_type
SalesOrderNumber	string
NULL	
SalesOrderLineNumber	int
NULL	
OrderDate	date
NULL	
CustomerName	string
NULL	
Email	string
NULL	
Item	string
NULL	
Quantity	int
NULL	
UnitPrice	float
NULL	
Tax	float
NULL	



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▶ Work with tables and SQL



```
1  %%sql
2  SELECT YEAR(OrderDate) AS OrderYear,
3         SUM((UnitPrice * Quantity) + Tax) AS GrossRevenue
4  FROM salesorders
5  GROUP BY YEAR(OrderDate)
6  ORDER BY OrderYear;
```

[156] - Command executed in 6 sec 12 ms by DAG on 6/15/2026, 12:27:53 PM

> Spark jobs (6 of 6 succeeded) Resources

...

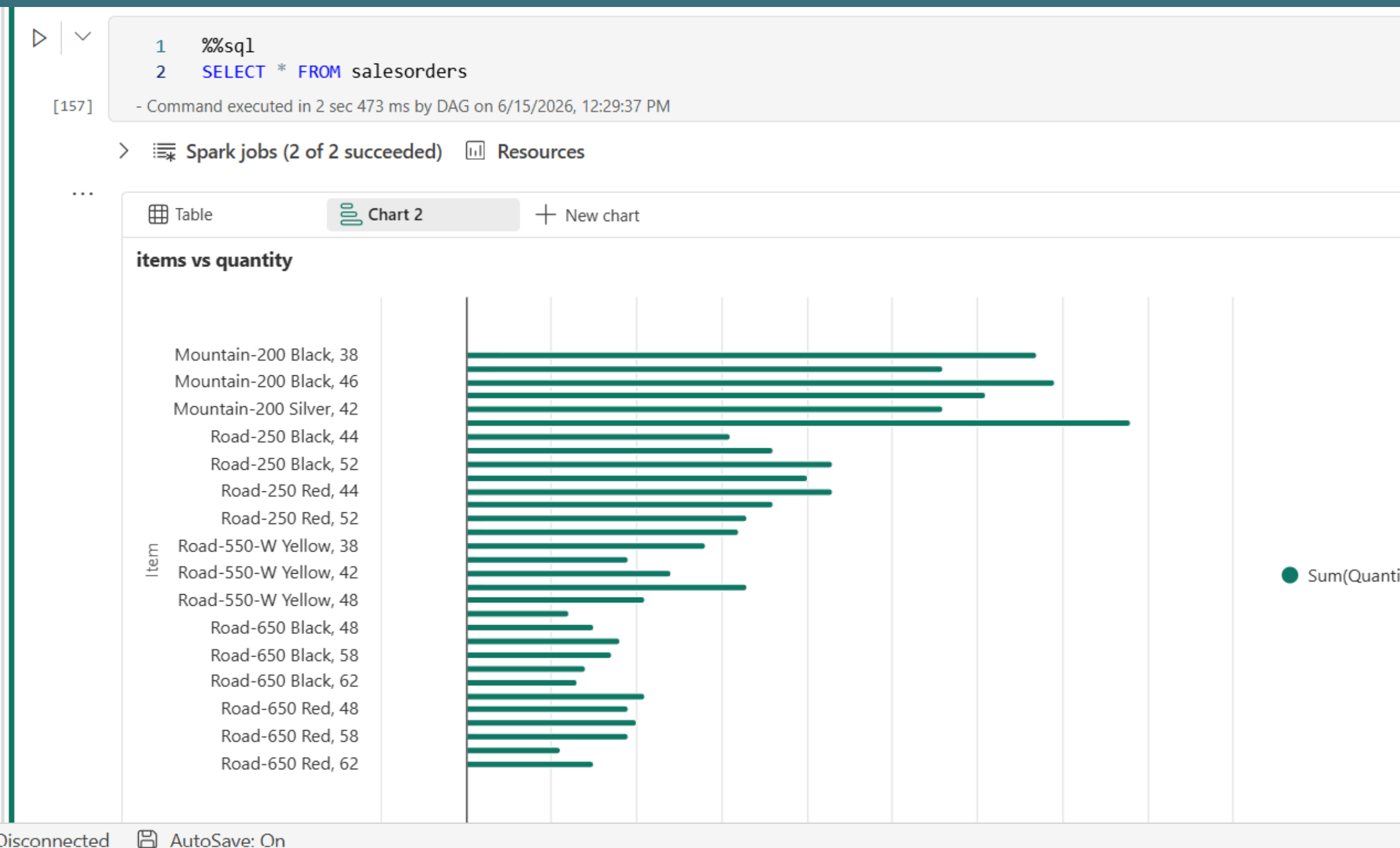
Table + New chart			
Table view			
	123 OrderYear	12 GrossRevenue	
1	2019	4172169.969970703	
2	2020	6882259.268127441	
3	2021	11547835.2916965...	



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Visualize data with Spark





DP-600 : Implement analytics solutions using Microsoft Fabric



Reference

